

Develop Real-time Applications with WebSockets and Socket.IO, Using Node.js



WebSockets is really useful for big applications, and your clients will definitely be impressed with the blazing speed it provides. Over time, it has improved considerably and got more stable. We might see a lot of scope for this technology very soon.



Real-time applications are those for which there is an instant response to a user request. Examples include your phone ringing when another person is calling you, or you receiving a text message from others, or even your MP3 player (in a way) using a real-time mechanism when you rock it to switch songs, forward or backward.

How do real-time applications work? First, let's discuss real-time Web applications only, as real-time systems are very complex and hard to describe.

Polling

Polling is the simplest real-time application mechanism. The mechanism operates as follows: after every regular interval, a request is sent to the server to check for updates in the form of a vote; the server then checks for any updates and sends an up-vote if updates are available and if not, it replies with a down-vote. At the user end, if the server casts an up-vote, the system further asks for the updated data which is then passed by the server. Once this process completes, again, requests are sent to the server at the regular intervals.

Ajax is preferred for polling as it is lightweight and can update parts of the page without refreshing the entire page. The problem is with the bandwidth, of which a large amount is used by both the server and the client. To reduce this, a method called *long-polling* has been introduced, whereby an HTTP request stays up until a new message is sent by the server. Gmail Chat is an example of this mechanism.

WebSockets

Some applications need very little or almost no latency in client-server communication. This is only possible when there is persistent communication between the browser and the server, as it works in a bi-directional communication between the server and the client. Eventually, you can perform all the polling tasks by using WebSockets with minimum latency. Another great feature that WebSockets allows is browser-to-browser communication. This means that two client browsers can communicate with each other without any latency, while the server processes and stores data.

Applications of WebSockets

WebSockets has become quite popular among Web developers these days, and can be used for the following applications:

- Chat application: Video and audio conferencing applications
- Multi-player online gaming
- Social feeds
- Simultaneous file sharing

There are endless applications that can be built with WebSockets, like stock market monitoring, sports updates, an online broadcast, etc. The question is how to use them? There are two incredibly great JavaScript libraries, namely, HTML5Rocks and Socket.IO. HTML5Rocks is natively available in modern browsers, while Socket.IO boasts of being the fastest among the competition. We personally prefer